

Fresh and Hardened Properties of SCC Incorporating Rice Husk Ash and Sugarcane Bagasse Ash

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Abstract. Self-compacting concrete represents a unique type of concrete that achieves self-compression through its mass, eliminating the need for external vibrations. The mechanical characteristics of self-compacting concrete can be enhanced by incorporating agricultural waste materials as partial substitutes for cement. In this experimental study, a graded M30 concrete mix was employed to produce self-compacting concrete by replacing cement with 6, 8, 10, 12, and 14% of sugarcane bagasse ash and rice husk ash. To assess the concrete's properties, both in its fresh and hardened states, several tests were conducted. Fresh properties were evaluated using l-box, slump flow, and U-box tests. The results indicated that sugarcane bagasse ash outperformed rice husk ash in enhancing the fresh properties of concrete up to a certain level of replacement. Hardened properties, including split tensile strength, compressive strength, and flexural strength, were tested at both 7-day and 28-day intervals. These tests revealed that rice husk ash exhibited superior performance in enhancing the hardened properties compared to sugarcane bagasse ash. In conclusion, this study underscores the suitability of both sugarcane bagasse ash and rice husk ash as viable options for partial cement replacement in self-compacting concrete, with each material exhibiting strengths in improving certain aspects of fresh and hardened concrete properties.

1 Introduction

Self-compacting concrete (SCC) is a type of concrete that can compact itself by its weight. It was first used in 1988 in Japan. Self-compacting concrete is used where the structure is very dense and it is very difficult to apply the vibrators. These days SCC is used where the need for high-performance concrete with a structure height between 12m to 15m is known as midrise structure. In this type of structure mainly grade M30 is used [1,2]. The current research is based on introducing SCC features into normal concrete mixes, with the incorporation of ash from agricultural wastes such as Sugarcane bagasse ash (BA) and Rice husk ash (RHA).

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After Brazil, India is the 2nd largest producer of sugarcane with a rough annual production capacity of 380 million tons, implying that considerable amounts of Bagasse Ash are produced and available for use. Previous SCC research suggests that Bagasse ash can be used in Self-compacting concrete manufacture and the approach allows for harmless Bagasse Ash disposal while also preventing pollution in the environment [1]. The surge in industrialization in India has resulted in a population increase, contributing to elevated food waste and the subsequent rise in the utilization of by-products like bagasse ash [2].

The fresh properties of a concrete mix are influenced by the use of chemical admixtures, such as superplasticizers and viscosity-modifying agents, commonly employed in standard concrete mixes. However, there is a noticeable absence of studies that investigate the incorporation of agricultural waste materials like Rice Husk Ash (RHA) and Bagasse Ash (BA) into normal self-compacting concrete to assess their impact on fresh properties. Sugarcane Bagasse ash and Rice husk ash operate as the VMA (Viscosity Modified Agent). This is due to the Viscosity Modified Agent admixture, which is low and costlier, thereby increasing the cost of manufacturing. SCC, BA, and RHA are all considered normal strength suited for replacing VMAs in SCC preparation. In light of the foregoing, the main focus of this paper is to observe the workability of self-compacting concrete through limited replacement of cement with sugarcane Bagasse ash and Rice husk ash to determine the best magnitudes of replacement.

The disposal challenge of sugarcane bagasse, a significant by-product of the sugar production industry, is exacerbated by its environmental impacts and health concerns [2].

The leading purpose of this study is to define the fresh and hardened properties of SCC with the fractional replacement of cement with RHA at different replacement levels to reduce the cost of concrete mix and use agricultural waste in concrete. The replacement is at 0, 5, 10, 15, and 20% by mass of the cement. While the RHA % increases, the fresh properties of concrete decrease in flowability, and the hardening properties are increased by 11.8% in compression at 10% and 7.8% in split tensile at the same 10% replacement level [3].

In a study about the partial replacement of BA as fine aggregate at the replacement level of 0-40%, the author conducted 7 days and 28-day tensile strength and compressive tests. The water-binder ratio is taken as 0.50, and the mix proportion is M15. He found that the slump increased by increasing Sugarcane bagasse ash, and the 28-days compressive strength and tensile strength increased by 7.90% and 14% at 10% addition of SCBA [4]. The author conducted a study in which he replaced the SCBA with fine aggregate of 0 to 40 levels, and he found that the flexural strength is improved by 14.41 at 10% of replacement, and the slump is reduced with the rise of sugarcane bagasse ash [5]. In another study, it is found at 10-30% replacement level, but the concrete decreases its flowability decrease with the increase of the % of SCBA [2].

In another research study, the replacement of cement was investigated within a range of 5% to 15%. Initially, this replacement resulted in a notable increase in compressive strength by 7.5%. However, as the replacement level reached 7.5%, there was a subsequent reduction in compressive strength, albeit still showing an overall improvement of 4.42% [6–8]. In a separate study, Sugarcane Bagasse Ash (SCBA) was used as a partial replacement in Self-Compacting Concrete (SSC) at various ratios, including 1%, 3%, 5%, 10%, 15%, 20%, and 25%. They conducted a study to determine the workability, mechanical strength, and microstructure of treated SCC. Authors explained that 10% use of SCBA with cement is the maximum share to induce economical SCC and utilization of SCBA which might cause greener self-compacting concrete [9]. The author investigated the use of admixture to reduce water from a concrete mix containing RHA as cement replacement. The author observed an increase in workability with the use of water reduction. Therefore, the content of the control

mixture is lesser than the superplasticizer content of RHA concrete mixtures. Additionally, the superplasticizer content can vary independently of the RHA proportion .

In another study replacing sand by sugarcane bagasse ash in concrete in two different proportions with the identical water-cement ratio. Mixtures are prepared with superplasticizer, and one mixture is prepared without superplasticizer. Due to this variation, a slump was observed. With increased content of SCBA, the flexural strength of the first mix rises. Therefore, the second mix of concrete flexural strength did not differentiate much from the normal concrete flexural strength [10]. In another study, the water absorption proportion accumulated among the existence of Sugarcane Bagasse Ash throughout twenty-eight days of activity, which is maybe because of the particular incontrovertible fact that Sugarcane Bagasse Ash is higher quality than OPC and is hygroscopic by nature. After ninety days of curing, the water absorption percentage is decreased by about 50%. As the result, the gradual closing of pores and permeable voids is decreased because of the accumulation of SCBA [11].

In another study when BA is used as a partial cement replacement and GS is used for natural sand, the workability of SCC is reduced, but fresh concrete density and air content are unaffected. Because of the coarse texture of GS and the increased porosity of BA, workability is projected to be reduced. The mechanical qualities of the concrete improve, including compression, split tension strength, and flexure strength. The largest increase was achieved by replacing 15% of the cement with BA and 10% of the FA with GS at the optimum concentration, and there is no effect on the modulus of elasticity [12]. Another study shows that the addition of SCBA reduced the diffusivity of chloride and increased its useful life by up to 97.3%. Pozzolan reaction and additional CSH formation reduced ASR with up to 5% SCBA addition [13]. In this study, SCC is prepared with the fractional replacement of cement with agricultural waste ash discussed. The consumption of RHA and BA compared to the fresh and hardened properties of concrete has been deliberated. The main aim remains to find a better material out of SCBA and RHA.

2 Properties of Material

2.1 Cement

OPC of grade 43, according to the Indian standard IS: 8112 (1989), is chosen for experimental purposes. The denoted specific gravity is 3.14. All tests are performed as per IS 4031 of 1988 [13-14].

2.2 Fine Aggregate

Aggregates procured from a 4.75mm sieve are considered for the research work. Fine aggregates with a fineness modulus of 3.05, specific gravity of 2.69, and water absorption of 1.1% are used for the mix.

2.3 Coarse Aggregates

Locally available coarse aggregates sized 10mm to 20mm are used. The specified coarse aggregates have a fineness modulus of 7.9 and a specific gravity of 2.74. Water absorption of 0.55% is observed.

2.4 Rice Husk Ash (RHA)

RHA is obtained from the incineration of rice husk, removed from the paddy during the milling process. The burning is carried out under controlled temperatures, maintained above 800 degrees Celsius. This process yields approximately 25%, with RHA composition being around 90% amorphous silica and 5% carbon content. The siliceous content in rice husk ash imparts pozzolanic properties to RHA. Silica reacts with Ca(OH)_2 , producing additional C-S-H, thereby enhancing concrete strength. RHA serves as an easily available raw material from agricultural waste for cement replacement.

2.5 Bagasse Ash

Bagasse ash is generated by burning sugarcane bagasse in sugar manufacturing factories or industries. Similar to fly ash, bagasse ash is black due to the presence of unburnt carbon. In contrast to cement, bagasse ash possesses a lower specific gravity, measured at 2.43. The particle size of bagasse ash is predominantly less than 85 microns, limiting its potential use and making landfill disposal the primary option. Bagasse ash is enriched with silica dioxide, promoting increased pozzolanic activity in concrete. Bagasse ash yields silica at about 80%, with a 73% absorption of moisture.

3 Experimental program

3.1 Sampling of Concrete

Various samples of self-compacting concrete are organized with a constant percentage of fly ash set at 35%, and then the cement is partially replaced with RHA and BA at varying percentages in two different mixes. The different proportions of materials used are shown in Table 3.2. The required number of samples for the strength test is provided in Table 3.1.

Table 3.1. Description of Specimens Used

Strength Test	Compressive Strength Test	Split Tensile Strength	Flexural Strength Test
Sample Type	Cubes	Cylinder	Beam
Sample Size	(150 X 150 X150)mm	(150 X300)mm	(100X 100 X500)mm
No. of Samples	66	66	66
Testing Days	7,28	7,28	7,28

Table 3.2. Percentage of Materials Used

Mix	Cement %	Fly ash %	Rice Husk Ash %	Bagasse Ash %	Fine Aggregate %	Coarse Aggregate %
R	65	35	0	0	100	100
BA6	61.1	35	6	0	100	100
BA8	59.8	35	8	0	100	100
BA10	58.5	35	10	0	100	100
BA12	57.2	35	12	0	100	100
BA14	55.9	35	14	0	100	100
RA6	61.1	35	0	6	100	100
RA8	59.8	35	0	8	100	100

RA10	58.5	35	0	10	100	100
RA12	57.2	35	0	12	100	100
RA14	55.9	35	0	14	100	100

3.2 Mix design

After conducting multiple trial mixes to attain the desired slump flow and achieve a 7-day strength target, the water-cement ratio was established at 0.45. The machine mixture is used to prepare the self-compacting concrete; first, the materials are dry-mixed for 2 to 3 minutes, and a high-range water-reducing agent of 2.65 kg/m3 is also used to attain the desired slump. The mixture design of concrete is done according to IS; 10262:2019 [17]. All other quantities of mixes are shown in Table 3.3. Then, the mixes are cast in a module and dried for 24 hours. After that, the casted samples are demolded and placed in water for curing.

Table 3.3. Design Mix Quantities of Material in kg/m³

Mix	Cement	Fly Ash	Rice Husk Ash	Bagasse Ash	Fine Aggregate	Coarse Aggregate	Water	Plasticizer
R	287	155	-	-	975	737	190	2.65
BA6	269.78	155	-	17.22	975	737	190	2.65
BA8	264.04	155	-	22.96	957	737	190	2.65
BA10	258.3	155	-	28.7	975	737	190	2.65
BA12	252.56	155	-	34.4	975	737	190	2.65
BA14	246.82	155	-	40.18	975	737	190	2.65
RA6	269.78	155	17.22	-	975	737	190	2.65
RA8	264.04	155	22.96	-	975	737	190	2.65
RA10	258.3	155	28.96	-	975	737	190	2.65
RA12	252.56	155	34.4	-	975	737	190	2.65
RA14	246.82	155	40.18	-	975	737	190	2.65

3.3 Methodology

The M30 grade concrete is commonly used in the construction industry. In this experimental study, a graded mix of M30 concrete has been used to prepare self-compacting concrete with cement replacement using SCBA and RHA. The prepared concrete mix is compared with normal SCC without any replacement. The concrete is prepared according to the proportions given in Table 3.2. The concrete is prepared by dry mixing the required materials in a machine mixture until a uniform mix is achieved. Then, water mixed with the required super-plasticizer is added to the mix, and concrete is prepared. Firstly, the prepared concrete is tested in a fresh state for the fresh properties of concrete using three methods: L-box, slump flow, and v-funnel tests. For the hardened properties of concrete, several samples of cubes, cylinders, and beams are cast for the split tensile strength, compressive strength, and flexural strength of concrete. Samples cast are opened after 24 hours and placed in a curing pond for

curing. The strength properties are checked on the 7th day and 28th days. At least three samples are tested for each strength property for each age of the test.

4 Result

4.1 Fresh Properties of SCC

4.1.1 Slump Flow

Slump flow assesses the flow ability of self-compacting concrete under gravitational action, minimizing external forces to zero for compaction effects. Previous studies in this field achieved a slump value of 550mm, whereas our well-precised test, conducted 2 to 3 seconds after mixing, falls under the T500 slump flow. The values are illustrated in Fig. 4.1.

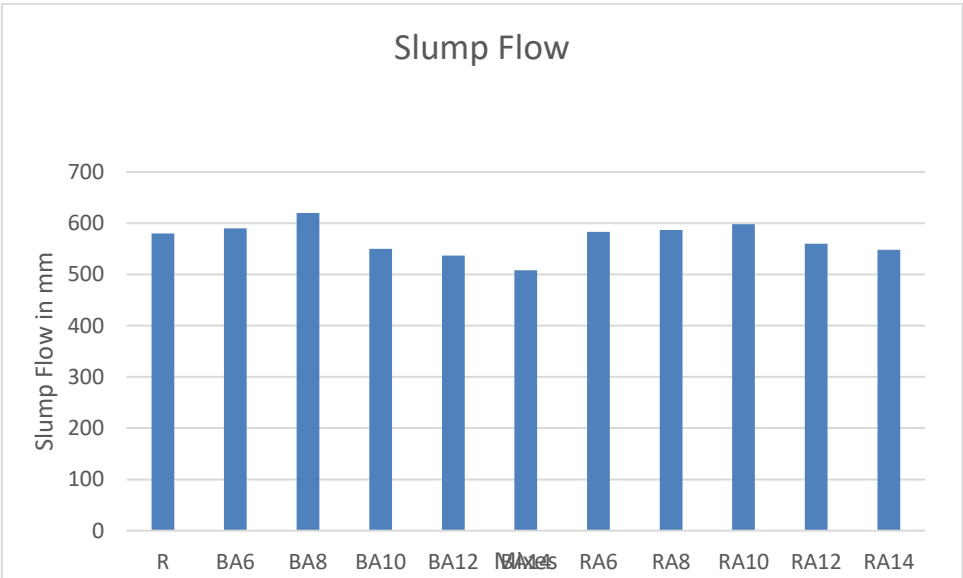


Fig. 4.1. Variation in Slump Flow Values

From the test results, it is found that normal concrete has a 580mm slump flow, and the BA6 sample increases slump flow values by 1.72%, while BA8 increases the slump value by 6.89% compared to normal concrete. However, it is observed that after increasing bagasse ash by 10%, 12%, and 14%, the slump flow values gradually decrease compared to normal concrete slump test values. Additionally, it is observed that after replacing cement with 8% and 10% of rice husk, the slump test results increase by 1.2% and 3.8%, respectively. Bagasse ash shows better properties than the normal mix and the rice husk ash replacement. Fig. 4.2 visually portrays the slump flow characteristics of self-compacting concrete, providing a valuable reference for assessing its flow ability and workability in construction scenarios.



Fig. 4.2. Slump Flow Test of SCC

4.1.2 L-box

The L-box test assesses the transient capability of self-compacting concrete by its weight without any vibration. This test defines the ability of concrete to pass through confined spaces and fine openings, validated with a misnamed ratio of the depth of concrete in the lateral section synchronous to the depth of concrete in the longitudinal desertion, which is also validated at 0.8. Findings reveal that the normal concrete L-Box ratio is 0.93, and the BA6 sample has a maximum L-box ratio of 0.96. With an increase in bagasse ash percentage replacing cement in SCC, the L-box ratio decreases gradually. Conversely, the RA6 sample shows a 0.92 L-box ratio, and increasing rice husk ash percentage instead of cement in SSC decreases the L-box ratio, as shown in Fig. 4.3.

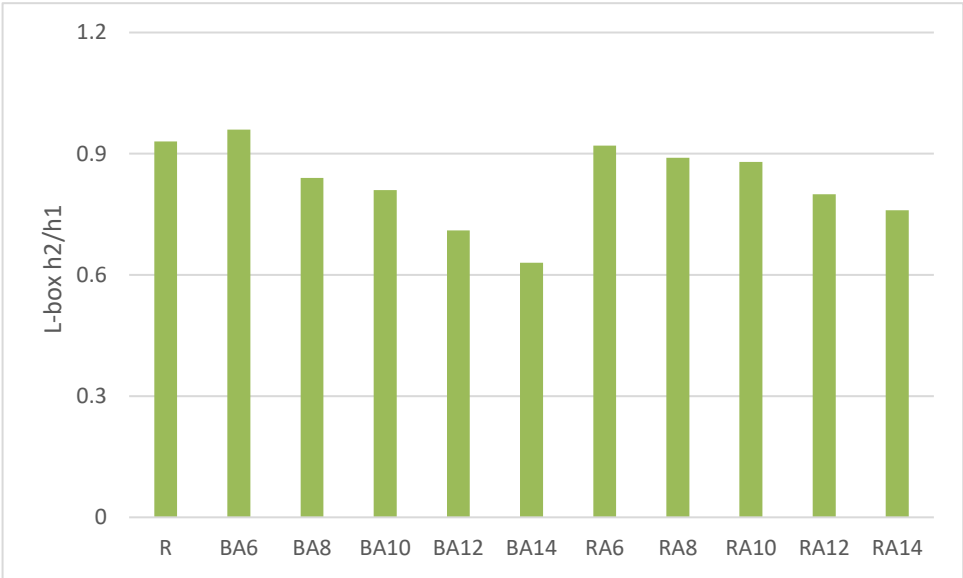


Fig. 4.3. Variation in Ratios of L-box

4.1.3 U-box

The U-box test evaluates the filling ability of self-compacting concrete without vibrations. Fig. 4.4 illustrates the variation in U-box test values conducted on SCC prepared with partial cement replacement with sugarcane bagasse ash and rice husk ash in different mixes. The minimum U-box value among the other mixes indicates good filling and passing ability. Normal concrete has a U-box value of 28mm. For bagasse ash, the BA8 sample shows the minimum U-box value at 25mm, and the BA14 sample has a maximum U-box value of 54mm. In contrast, using RA6 sample showed a 26mm U-box value, and the RA14 sample showed a maximum U-box value of 41mm.

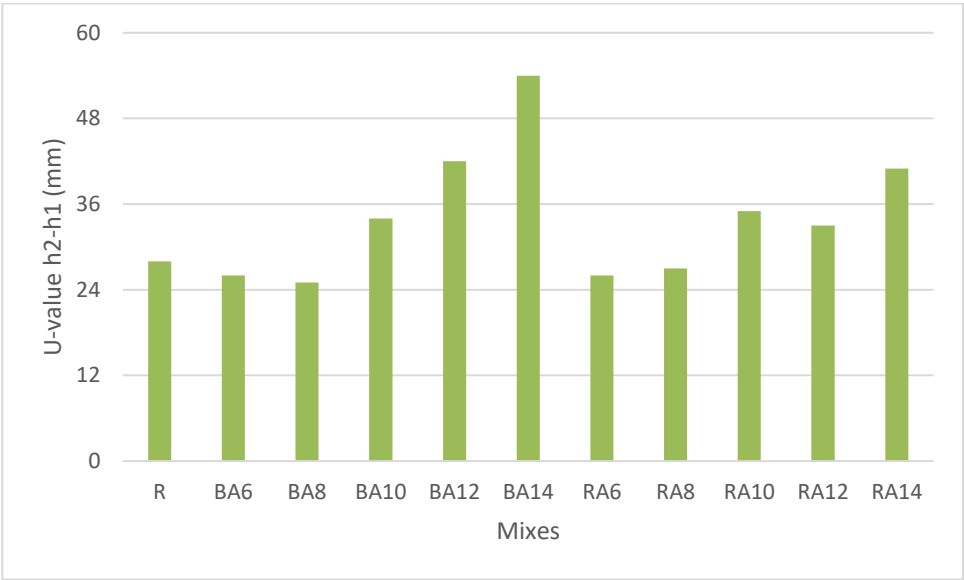


Fig. 4.4. Variation in U-box Values

4.2 Hardened Properties

4.2.1 Compressive Strength

Various cube specimens of size 150X150X150mm are prepared for the compressive strength of concrete. For every mix, three samples are cast for the 7th-day strength, and three are for the 28th-day strength of concrete. Compressive strength is determined with the help of a CTM under a loading rate of 13.7N/mm2 according to BIS: 516–1999 (1959). It was observed that when we increase the percentage of sugarcane bagasse ash (SCBA), the compressive strength increases up to a certain level, as shown in Fig. 4.4. The best result is found at 8% of SCBA, which is 33.97MPa; after that, when we increase the percentage of SCBA, the compressive strength decreases. A similar trend is observed with Rice Husk Ash (RHA). RHA increases the compressive strength up to 10% substitution, reaching 35.79MPa; however, further increases in RHA result in a decrease in compressive strength, as shown in Fig. 4.5. The outcomes indicate that Rice Husk Ash provides better results than SCBA.

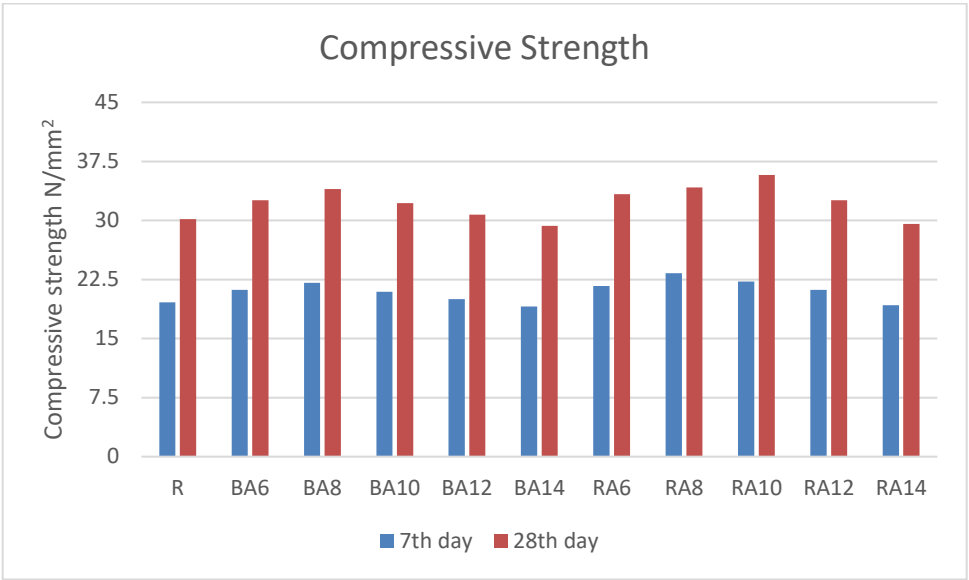


Fig. 4.5. Compressive Strength Variation

In Fig. 4.6 of this research paper, the compressive strength apparatus for self-compacting concrete is depicted, offering a clear visualization of the testing setup. This illustration serves as a crucial reference point, facilitating a better understanding of the experimental methodology employed to evaluate the strength properties of self-compacting concrete in the study.



Fig. 4.6. Compressive Strength Testing

4.2.2 Flexural Strength

Various beam samples of size 100X100X500mm are prepared for the flexural strength of concrete. For every mix, three beams for 7 days and three beams for 28 days are cast for flexural strength testing. The flexural test is conducted with the help of a flexural strength testing machine with two-point loading. It was observed that the highest strength is witnessed at 8% substitution of cement with SCBA, reaching 4.45 MPa. With an increase in the percentage of SCBA, the flexural strength decreases. Similarly, the highest strength with fractional replacement of RHA is at the 10% substitution level, reaching 4.46 MPa; however, further increases in RHA result in a decrease in flexural strength. RHA and SCBA show similar results in flexural strength, with RHA providing slightly more strength, as shown in Fig. 4.7. Fig. 4.8 in the research paper showcases the flexural strength testing for concrete, providing a visual insight into the experimental setup employed to assess the material's ability to withstand bending forces.

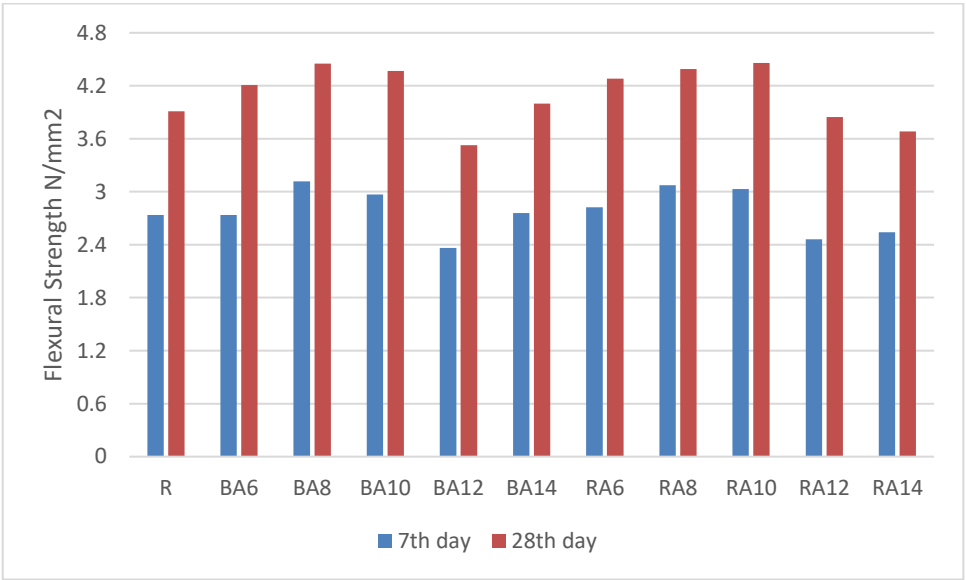


Fig 4.7. Flexural Strength of different Mix Design



Fig. 4.8. Flexural strength of different mix design

4.2.3 Split Tensile Strength

Various cylinder samples are cast of size 150mmX300mm for determining the split tensile strength. For every mix, three cylinders for 7 days and three cylinders for 28 days are cast for split tensile strength testing. The experiment is performed on a compression testing machine with a loading rate of 13.7N/mm2 according to BIS: 516–1999 (1959).

It was observed that the highest strength is detected at 8% of replacement of cement with SCBA, reaching 3.59 MPa at 28 days. With an increase in the percentage of SCBA, the split tensile strength decreases. Similarly, the highest strength with fractional replacement of RHA is at the 10% replacement level, reaching 3.68 MPa at 28 days; however, further increases in RHA result in a decrease in split tensile strength, as shown in Fig. 4.9. Fig. 4.9 in the research paper illustrates the split tensile strength testing of concrete, visually presenting the methodology employed to evaluate the material's ability to withstand tensile forces through axial splitting.

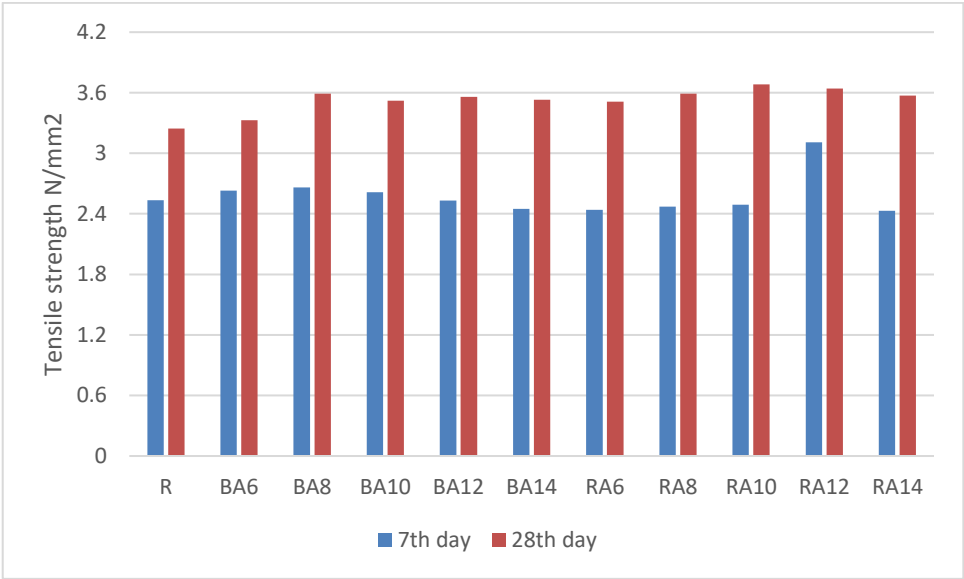


Fig. 4.9. Split Tensile Strength Variation

5 Conclusion

This experimental study aimed to assess the potential of utilizing agricultural waste materials, specifically sugarcane bagasse ash and rice husk ash, as partial substitutes for cement in self-compacting concrete (SCC). SCC, renowned for its self-compression capabilities, is an innovative construction material, and this research sought to enhance its mechanical properties through sustainable means. The following conclusions are derived from the present study:
The highest compressive strength values are achieved when 10% of the cement is replaced with Rice Husk Ash (RHA). However, increasing the percentage of RHA results in a decrease in compressive strength, as depicted in Fig. 4.5.

In the assessment of flexural strength, it was observed that Rice Husk Ash (RHA) outperformed Sugarcane Bagasse Ash (BA) when 10% of the cement was partially substituted. The maximum flexural strength recorded was 4.46 MPa, as illustrated in Fig. 4.7. The split tensile strength test indicates that rice husk ash performs better than sugarcane bagasse ash. The highest value of split tensile strength is at the 10% partially substitution level of cement by rice husk ash, which was 3.68 MPa, as shown in Fig. 4.9. In the assessment of fresh properties, it was found that SCBA gives better results than rice husk ash. The highest value of slump flow is 620mm, as clearly shown in Fig. 4.1.

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